1. (10 marks)

   State whether each of the following statements is true or false. Give a brief justification for your answer. You do not need to provide fully detailed proofs, and simply a counterexample without proof is sufficient.

   (a) True: it is the discrete topology.

   (b) False. Sierpinski's space. Let A be the open set $\{0\}$.

   (c) False. $\mathbb{N}$ and $\mathbb{Z}$ are homeomorphic and the inclusion function from $\mathbb{N}$ into $\mathbb{Z}$ is an embedding.

   (d) False. Singletons are not contained in basic open sets as $\mathbb{N}$ is infinite.

   (e) True. Hausdorff spaces are T_1.

   (f) False. The co-finite topology on $\mathbb{N}$.

   (g) False. The comb space

   (h) False. $p : \mathbb{R} \to \mathbb{S}^1$. A path in $\pi((0,1), \mathbb{S}^1$ can be lifted to start at n, for any integer n.

   (i) True. They are homeomorphic spaces.

   (j) False. It is the non-abelian free group on 2 generators.

2. (15 marks)

   (a) i. Let $\mathcal{U}$ be an open cover of X. Let $U \in \mathcal{U}$. Then $X \setminus U$ is finite and so $X \setminus U$ can be covered by finitely many sets in $\mathcal{U}$.

   ii. Let $a, b \in X$, $a \neq b$. Then $\{a\}$ and $\{b\}$ are disjoint closed sets. If U is an open set containing a, then $X \setminus U$ is finite. Hence, $X \setminus U$ can't contain an open neighbourhood of b disjoint from $\{a\}$.

   (b) Let $A = \{x \in X : f(x) \neq g(x)\}$, and suppose $x \in A$. Since $f(x) \neq g(x)$, there are open sets $U, V \subset Y$ such that $f(x) \in U$, $g(x) \in V$ and $U \cap V = \emptyset$. Let $W = f^{-1}(U) \cap g^{-1}(V)$. Then W is open and $x \in W$. Moreover, $W \subset A$. Thus A is open, so $\{x \in X : f(x) = g(x)\}$ is closed.

   (c) Suppose X and Y are first countable spaces. For each $x \in X$, let $\mathcal{N}_x$ be a countable neighbourhood open basis at x, and for each $y \in Y$, let $\mathcal{M}_y$ be a countable open neighbourhood basis at y.

   Let $\mathcal{B} = \{U \times V : U \in \mathcal{N}_x, V \in \mathcal{M}_y\}$. If $(x, y) \in X \times Y$ and W is a neighbourhood of (x, y), then there is a basic open set $A \times B$ such that $(x, y) \in A \times B \subseteq X \times Y$. It now follows that there is a set $U \times V \in \mathcal{B}$ such that $(x, y) \in U \times V \subseteq A \times B$.

3. (15 marks)

   (a) Prove that if X and Y are connected spaces then $X \times Y$ is connected.

   Proof Suppose that (x_1, y_1) and (x_2, y_2) are arbitrary distinct points in $X \times Y$. Clearly $\{x_1\} \times Y$ is homeomorphic to Y (the bijection defined $(x_1, y) \mapsto y$ is a

homeomorphism) and hence is a connected subspace of $X \times Y$. Similarly, $X \times \{y_2\}$ is homeomorphic to X and hence is a connected subspace of $X \times Y$. Observe that

$$(X \times \{y_2\}) \cap (\{x_1\} \times Y) = \{(x_1, y_2)\} \neq \emptyset,$$

so $(X \times \{y_2\}) \cap (\{x_1\} \times Y)$ is connected and hence (x_1, y_1) and (x_2, y_2) are connected in $X \times Y$.

OR

Choose a point $b \in X$. Clearly $\{b\} \times Y$ is homeomorphic to Y (the bijection defined $(b, y) \mapsto y$ is a homeomorphism) and hence is a connected subspace of $X \times Y$. Similarly for every $y \in Y$, $X \times \{y\}$ is homeomorphic to X and hence is a connected subspace of $X \times Y$. Observe that for every $y \in Y$,

$$(X \times \{y\}) \cap (\{b\} \times Y) = \{(b, y)\} \neq \emptyset,$$

and hence, by Theorem 5.6,

$$X \times Y = \bigcup_{y \in Y} X \times \{y\}$$

is connected.

(b)

($\Rightarrow$) Let $f : X \to [0, 1]$ be continuous with $f(A) = 0$ and $f(B) = 1$. If $r < 0$ let $U_r = \emptyset$ and if $r > 1$ let $U_r = X$. For each $r \in [0, 1]$ let

$$U_r = \left\{ x \in X : f(x) < \frac{r+1}{2} \right\}.$$

Then $\{U_r : r \in \mathbb{R}\}$ is a Urysohn family of the required form. Note that if $r < 0$ then $U_r = \emptyset$, and if $r > 1$ then $U_r = X$.

($\Leftarrow$) Let $\mathcal{U} = \{U_r : r \in D\}$ be a Urysohn family with $A \subset U_0$ and $B \subset X - U_1$, and let $\phi = \phi_{\mathcal{U}} : X \to \mathbb{R}$ be as in Lemma 0.2. Define $f : X \to [0, 1]$ by

$$f(x) = \begin{cases} 0 & : \quad \text{if} \quad \phi(x) \leq 0 \\ \phi(x) & : \quad \text{if} \quad \phi(x) \in [0, 1] \\ 1 & : \quad \text{if} \quad \phi(x) \geq 1. \end{cases}$$

By Theorem **??** f is continuous. Furthermore, if $x \in A$ then for each $r > 0$, $x \in U_r$; so $\phi(x) \leq 0$ and hence $f(x) = 0$. On the other hand, if $x \in B$ then for each $r < 1$, $x \notin U_r$, so $\phi(x) \geq 1$ and hence $f(x) = 1$.

4. (15 marks)

(a) $E = [0, 1]$ with p the identity. Take any discrete set D, let $E = [0, 1] \times D$ and define p to be the natural projection.

(b)

$$H_q^{\Delta}(X) = \begin{cases} \mathbb{Z} & \text{if} \quad q = 0, 1 \\ 0 & \text{if} \quad q > 1 \end{cases}$$

With appropriate working.

Appendix

Definition 0.1. Let X be a space. By a *Urysohn family* in X we mean a family $\{U_r : r \in D\}$ of open subsets of X satisfying:

(i) $\bar{D} = \mathbb{R}$;

(ii) $\bigcup_{r\in D} U_r = X$ and $\bigcap_{r\in D} U_r = \emptyset$;

(iii) if $r < s$ then $\overline{U_r} \subset U_s$.

Lemma 0.2. *Let $\mathcal{U} = \{U_r : r \in D\}$ be a Urysohn family in the space X and define a function $\phi_{\mathcal{U}} : X \to \mathbb{R}$ by*

$$\phi_{\mathcal{U}}(x) = \inf\{r \in D : x \in U_r\}.$$

Then $\phi_{\mathcal{U}}$ is continuous.

Lemma 0.3. *[750] Let A and B be disjoint closed subsets of a space X. Then A and B are completely separated in X iff there is a Urysohn family $\{U_r : r \in D\}$ with $0, 1 \in D$, $A \subseteq U_0$ and $B \subseteq X \setminus U_1$.*